

Practice Quiz 09D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 09

Well-definedness on a quotient; onto maps and inverses.

Write $[x]_9$ for the congruence class of x modulo 9.

1. Consider

$$f : \mathbf{Z}_9 \longrightarrow \mathbf{Z}_9, \quad f([x]_9) = [2x + 4]_9.$$

(a) Prove that f is well defined.

Solution Suppose $[x]_9 = [y]_9$. Then $9 \mid (x - y)$, so $x - y = 9k$ for some integer k . Therefore

$$(2x + 4) - (2y + 4) = 2(x - y) = 9(2k).$$

Thus $9 \mid ((2x + 4) - (2y + 4))$, so $[2x + 4]_9 = [2y + 4]_9$. Hence the output does not depend on the representative and f is well defined. ■

(b) Prove that f is onto and determine f^{-1} .

Solution The number 5 is an inverse of 2 modulo 9, since $2 \cdot 5 \equiv 1 \pmod{9}$. Given an arbitrary $[y]_9$, choose

$$[x]_9 = [5(y - 4)]_9.$$

Then

$$f([x]_9) = [2 \cdot 5(y - 4) + 4]_9 = [y]_9,$$

because $2 \cdot 5 \equiv 1 \pmod{9}$. Thus f is onto. The calculation also gives

$$f^{-1}([y]_9) = [5(y - 4)]_9.$$

The same representative check as in part (a), with coefficient 5, shows that this inverse rule is well defined. ■